Complements to 10: Making Ten

Addition Strategies to 10 • Standard: CCSS.MATH.1.OA.C.6

BOOK STORY OPENER: Making a Ten First (8 + 5 = 10 + 3 = 13) & The Double-Height '10' Carry Block Trade

Why is 10 + 3 so much easier for our brains than 8 + 5? Because base-10 place value groups everything into full tens! In Chapter 3 of From Blocks to the Stack, students slide 2 dots from the right tray into the 2 empty pockets of the left tray to complete a full 10, then physically swap all 10 loose dots for one double-height '10' Carry Block.

Companion Book: From Blocks to the Stack (SC-6 Manual) — Ch. 3: Making a Ten First & Trading 10 Ones

1. CONCRETE MANIPULATIVE SETUP

Manipulative Kit: Dual Ten-Frame Trays & Counters

Place 7 unnumbered counters into the Ten-Frame: fill the top row of 5, then place 2 in the bottom row.

2. STUDENT PREDICTION (ASK BEFORE PRESSING KEYS)

Look at the empty wells without counting one-by-one. 3 empty spaces mean 7 + 3 = 10.

3. CALCULATOR ACTION (SC-6 KEYCAPS & STACK TRACE)

Keystroke Flow: [1] [0] [ENTER] [7] [-]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[1]	1	Enter 1
[0]	10	Full frame capacity (10) in X
[ENTER]	10	10 lifted to Y
[7]	7	Occupied counters (7) in X
[-]	3	10 - 7 = 3: Exactly 3 empty wells to make 10!

4. MATHEMATICAL EXPLANATION & RPN ADVANTAGE

The ten-frame physically structures 10 as two groups of 5. Subtraction verifies the visual complement.

Core Investigation Problem:

A ten-frame has 6 counters. How many more counters are needed to make 10?

Bridging Through 10: The Make-Ten & Carry Strategy

Addition Within 20 • Standard: CCSS.MATH.1.OA.C.6

1. FOLLOW-UP EXTENSION SETUP & PREDICTION

Setup: Place 8 unnumbered counters in Ten-Frame 1. Prepare 5 counters beside the board. Prediction: Take 2 from the 5 to fill Ten-Frame 1 to 10. Trade the full frame for one '10' block in the Carry Bay! 3 loose counters remain. 10 + 3 = 13.

2. EXTENSION KEYSTROKE SEQUENCE & STACK TRACE

[8] [ENTER] [2] [+] [3] [+]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect
[8]	8	8 counters on frame
[ENTER]	8	8 lifted to Y
[2]	2	2 counters borrowed to make 10
[+]	10	8 + 2 = 10 (Full frame traded for '10' carry block!)
[3]	3	Remaining 3 counters
[+]	13	10 + 3 = 13 total!

3. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Counting on by ones across 10 (e.g., 8 + 5 as 9, 10, 11, 12, 13) instead of decomposing 5 into 2 + 3 to complete the Ten-Frame.

Hands-On Block Remediation: Have the student physically slide 2 counters from the second group into the 2 empty wells of the first Ten-Frame, then key [8] [ENTER] [2] [+] [3] [+].

Misconception 2: Reading teen numbers as two separate single digits (seeing 14 as 1 and 4 rather than 1 full Ten-Frame plus 4 ones).

Hands-On Block Remediation: Snap a full 10-tray beside 4 loose counters and key [10] [ENTER] [4] [+] so students see 14 formed from 10 in Register Y and 4 in Register X.

4. EXTENSION MATHEMATICAL INSIGHT

Bridging decomposes 8 + 5 into (8 + 2) + 3. On RPN, intermediate sums remain in the accumulator register.

Ten-Frames: Complements to 10 & 5-Structure

Name: _____ Date: _____ Class: _____

STUDENT REFERENCE GUIDE: TEN-FRAME 5-STRUCTURE & CARRY RULES

- The 2x5 grid holds unnumbered counter blocks. Spatial position builds visual subitizing without numeral distraction.
- Subitizing: 7 is seen immediately as 5 on top plus 2 on bottom.
- Dual 10-Frame Trays: Students use two dedicated 10-frame trays (strictly 10 slots each, solid bottoms) side-by-side on the desk. The 12-slot main storage bin keeps 10-bundles and the 2 carry blocks separate.

Key Sequence: `1` `0` `ENTER` `7` `-`

Problem 1: What is the complement to 10 for 4? Write the keystrokes.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on SC-6

Problem 2: Solve 7 + 5 using Make-Ten: first make 10, then add 2.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on SC-6

Problem 3: A ten-frame has 9 counters. How many to make 10?

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on SC-6

Problem 4: Solve 8 + 6 using Make-Ten on the calculator.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on SC-6

Unit 03: Ten-Frames: Complements to 10 & 5-Structure — Answer Key

Complete solutions and SC-6 verification keystrokes for Grade 1

1. CORE & EXTENSION INVESTIGATION SOLUTIONS

Core Investigation: A ten-frame has 6 counters. How many more counters are needed to make 10?

Core Solution: 4 counters. Keystrokes: [1][0] [ENTER] [6] [-]

Core Check Keys: 1 0 ENTER 6 -

Extension Challenge: Use the Make-Ten strategy to solve 9 + 6 on your calculator.

Extension Solution: 15. Keystrokes: [9] [ENTER] [1] [+] [5] [+]

Ext. Check Keys: 9 ENTER 1 + 5 +

2. STUDENT LAB WORKSHEET SOLUTIONS (PROBLEMS 1–4)

Problem 1: What is the complement to 10 for 4? Write the keystrokes.

Expected Answer: 6

SC-6 Verification Keys: 1 0 ENTER 4 -

Problem 2: Solve 7 + 5 using Make-Ten: first make 10, then add 2.

Expected Answer: 12

SC-6 Verification Keys: 7 ENTER 3 + 2 +

Problem 3: A ten-frame has 9 counters. How many to make 10?

Expected Answer: 1

SC-6 Verification Keys: 1 0 ENTER 9 -

Problem 4: Solve 8 + 6 using Make-Ten on the calculator.

Expected Answer: 14

SC-6 Verification Keys: 8 ENTER 2 + 4 +

Hands-On Block & Stack Verification Protocol

Students first build the physical block model and write their mathematical strategy and prediction before verifying their result on the StackCalc calculator.